

Manh Nguyen

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Aspiring **Software Engineer** with 2+ years of experience developing **full-stack applications** and **AI-powered solutions**. Skilled in **full-stack development**, **RAG pipeline implementation**, and **LLM orchestration**, with a focus on building **user-centric scalable products**. Passionate to leverage into real-world problems and contribute to innovative teams.

EXPERIENCE

AI Full Stack Developer

Jul. 2024 – Oct. 2025

Home Invease

- Developed an **AI-powered household inventory management application** enabling users to track and organize items through **object recognition** and **receipt scanning**.
- Designed **Spring Boot microservices** with **RESTful APIs** to orchestrate inventory management, AI inference, and secure authentication across mobile and web applications.
- Architected **RAG pipeline** combining **ChromaDB vector search** with **Ollama LLM models** to enhance context-aware inventory search, while reducing query latency by **30%**.
- Fine-tuned **image classification model** to recognize household items from photos, paired it with **Tesseract OCR** for receipt scanning, then fed the raw text into the LLM model, eliminating manual entry.
- Collaborated across the team from **system design** to **testing**, **deployment**, and iterative feature delivery—using **Git-based workflows** and **Microsoft Azure cloud services**.

PROJECTS

Link AI Assistant | *React, Next.js, Express.js, TypeScript, Claude, OAuth2*

[GitHub](#)

- Architected an **AI-powered file management assistant** integrating **Google Drive API** to automate file organization and storage optimization.
- Designed **weighted scoring engine** combining metadata features to calculate **confidence scores** and **relationship** between files.
- Developed an integrated **Claude Code pipeline** that ranks file relationships and provides organization suggestions based on **similarity confidence scores**.
- Established **CI/CD pipelines** and deployed as website and browser extension using **Chrome Extension Manifest V3** and **GitHub Actions**.

Loan Default Prediction Model | *Python, PostgreSQL, Apache Spark, Machine Learning*

[GitHub](#)

- Engineered an **end-to-end ML decision-support platform** using **Apache Spark** trained on over **850,000 customer records**.
- Integrated **data preprocessing**, **predictive modelling**, and **RESTful API services** for automated credit risk assessment.
- Built a **feature engineering pipeline** generating financial risk indicators to improve prediction reliability.

EDUCATION

University of Wollongong

Wollongong, NSW

Bachelor of Computer Science (Artificial Intelligence and Big Data)

Oct. 2022 – Oct. 2025

- Distinction Grade:** Advanced Programming, Database Systems, Modern Artificial Intelligence, AI & Machine Learning, Big Data Management, Big Data Mining Technologies.

TECHNICAL SKILLS

Languages: Java, Python, C++, C#, TypeScript, JavaScript, SQL, R

Software Engineering: Spring Boot, React, Next.js, Node.js, RESTful APIs, PostgreSQL, MongoDB, Firebase, ChromaDB, OAuth2, JWT

AI & Big Data: LLMs, RAG, NLP, Computer Vision, TensorFlow, PyTorch, Ollama, Apache Spark, Hadoop

Cloud & DevOps: Microsoft Azure, AWS, Google Cloud Platform, Docker, Git, GitHub Actions